

Part A — Fundamentals

1. What is Data Analysis?

Definition

Data Analysis is the process of collecting, cleaning, transforming, and interpreting data to extract useful information and support decision-making.

Purpose

The main purpose of data analysis is to discover patterns, trends, and insights from raw data that can help organizations make informed decisions.

Types of Data Analysis

1. Descriptive Analysis

Describes what happened in the past.

Example:

- Monthly sales reports
- Customer count reports

2. Diagnostic Analysis

Explains why something happened.

Example:

- Finding reasons behind customer churn.

3. Predictive Analysis

Predicts future outcomes using historical data.

Example:

- Predicting whether a customer will churn.

4. Prescriptive Analysis

Suggests actions to achieve desired outcomes.

Example:

- Recommending offers to retain customers.

Importance in Decision Making

- Helps businesses make data-driven decisions.
 - Identifies trends and patterns.
 - Improves operational efficiency.
 - Supports strategic planning.
 - Reduces risks and uncertainties.
 - Enhances customer satisfaction.
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2. Steps of a Data Science Project

A Data Science project follows a systematic lifecycle from problem identification to deployment.

1. Problem Understanding

Understand the business problem and define project objectives.

Example: Predict whether a customer will churn.

2. Data Collection

Gather data from various sources such as:

- CSV files
- Databases
- APIs
- Websites

3. Data Cleaning

Improve data quality by:

- Handling missing values
- Removing duplicates
- Correcting inconsistencies
- Treating outliers

4. Exploratory Data Analysis (EDA)

Analyze and visualize data to discover patterns and relationships.

Techniques:

- Histograms
- Boxplots
- Scatter plots
- Correlation heatmaps

5. Feature Engineering

Create or modify features to improve model performance.

Example:

$\text{TotalSpending} = \text{PurchaseAmount} \times \text{Quantity}$

6. Model Building

Train Machine Learning algorithms using prepared data.

Examples:

- Logistic Regression
- Decision Tree
- Random Forest

7. Model Evaluation

Measure model performance using metrics such as:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

8. Deployment

Deploy the trained model into a real-world environment.

Examples:

- Web Application
- Mobile Application
- Cloud Platform

9. Monitoring

Continuously monitor model performance after deployment.

Activities:

- Track accuracy
 - Detect data drift
 - Retrain when necessary
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3. Machine Learning Problem Statement

Problem Statement

The objective of this project is to predict whether a customer will churn based on customer demographics, purchase behavior, membership details, and customer satisfaction data.

Input Features

The model uses the following input features:

- Age
- Gender
- Income
- MembershipType
- PurchaseAmount
- Quantity
- CustomerSatisfaction
- WebsiteVisitsPerMonth
- LastPurchaseDaysAgo
- EmailSubscribed
- PreferredPaymentMethod
- City
- State

Target Variable

Churn

Value	Meaning
0	Customer Stays
1	Customer Churns

Problem Type

Supervised Machine Learning

The dataset contains labeled examples with known churn outcomes.

Classification Problem

The output belongs to one of two categories:

- Churn = Yes
- Churn = No

Therefore, this is a **Binary Classification Problem**.

Expected Outcome

The model should predict whether a customer is likely to leave the company so that businesses can take proactive retention measures.

4. Tensors

What are Tensors?

A tensor is a mathematical structure used to represent numerical data in multiple dimensions.

Tensors are the fundamental building blocks of Machine Learning and Deep Learning frameworks such as TensorFlow and PyTorch.

A tensor can be viewed as a generalized form of scalars, vectors, and matrices.

Tensor Dimensions

0D Tensor (Scalar)

A scalar contains a single value.

Example:

```
import numpy as np
```

```
scalar = np.array(10)
print(scalar.ndim)
```

Output:

```
0
```

1D Tensor (Vector)

A vector is a collection of values arranged in one dimension.

Example:

```
vector = np.array([10,20,30,40])
print(vector.ndim)
```

Output:

```
1
```

2D Tensor (Matrix)

A matrix consists of rows and columns.

Example:

```
matrix = np.array([
    [1,2,3],
    [4,5,6]
])
print(matrix.ndim)
```

Output:

```
2
```

3D Tensor

A 3D tensor is a collection of matrices.

Example:

```
tensor3d = np.array([
    [[1,2],[3,4]],
    [[5,6],[7,8]]
])

print(tensor3d.ndim)
```

Output:

```
3
```

Tensor Representation

Tensor Type	Dimension	Example
Scalar	0D	10
Vector	1D	[10,20,30]
Matrix	2D	[[1,2],[3,4]]
3D Tensor	3D	Collection of matrices

Importance of Tensors

- Store multidimensional data efficiently.
 - Fundamental data structure in Deep Learning.
 - Used in TensorFlow and PyTorch.
 - Support fast mathematical computations.
 - Enable processing of images, videos, text, and numerical datasets.
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Conclusion

Data Analysis and Machine Learning play a crucial role in extracting meaningful insights from data and solving real-world problems. The Customer Churn Prediction project follows the complete Data Science lifecycle, from data collection and preprocessing to model building and evaluation. Tensors provide the underlying structure for handling multidimensional data used in Machine Learning and Deep Learning applications.